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Why We Should Study the Broader Autism Phenotype in Typically Developing Populations

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The broader autism phenotype (BAP) is a term applied to individuals with personality and cognitive traits that are similar to but milder than those observed in autism spectrum disorder (ASD). Subtle autistic traits in the core diagnostic domains of social communication and rigid behavior were described in family members of people with an ASD even in the initial reports of ASD. In this article, we discuss the benefits and limitations of researching the BAP in typically developing individuals for understanding autism and development.

In this article, we propose that the study of the broader autism phenotype (BAP) among typically developing persons can inform our conceptual understanding of both autism spectrum disorder (ASD) and typical development. The BAP is a subclinical presentation of one or more behaviors or traits that are qualitatively similar to features of autism. These autistic-like behaviors were initially described in family members of children with autism, and researchers have long examined the BAP to uncover the genetic underpinnings of ASD. Here, we will review the concept and measurement of the BAP and outline the benefits to ASD research of this complementary model as well as the benefits to research in typical development.

ASD is a developmental condition that emerges in early childhood. It is characterized by impairments in social communication and the presence of restricted interests and repetitive behaviors. According to contemporary thought, ASD is a heterogeneous condition resulting from multiple possible etiological underpinnings that remain poorly understood (Betancur, 2011; Bruining et al., 2010; Jeste & Geschwind, 2014; Lenroot & KaYeung, 2013). As a consequence of this heterogeneity, no two people with an ASD are exactly alike in their presentation. What all people with an ASD do have in common is that their symptoms are clinically relevant and manifest functional limitations in their quality of life. However, the meaning of the “S” in ASD—the spectrum—highlights that the autistic traits are not only heterogeneous in nature but also vary in severity and extend to levels of nonclinical significance in the general population.

In his initial report describing 11 cases of autism, Kanner (1943) reported that many of the parents of his patients were “highly intelligent” people with extraordinary educational and professional attributes. At the same time, he saw subtle signs of some of his patients’ traits in these parents, including an obsession with details, social awkwardness, and rigid behaviors.

Subsequent reports of ASD echoed Kanner's accounts of these shared characteristics in family members (Creak & Ini, 1960; Eisenberg, 1957; Gillberg, 1989; Piven et al., 1994; Wolff, Narayan, & Moyes, 1988). In the era of Harlow's studies of social deprivation in monkeys (e.g., Harlow & Harlow, 1962) and Bowlby's research on mother-child attachment (e.g., Bowlby, 1958), the presence of these shared characteristics was misinterpreted as evidence of a possible environmental cause. Within this context, the observations of social aloofness and rigidity, particularly of the mother, gave rise to the "refrigerator mother" theory of autism (Bettelheim, 1967; Kanner, 1949). These earlier descriptions of shared characteristics among family members are remarkably similar in content to present-day notions of what characterizes the BAP.

The prominent role of genetics, rather than parenting, in both the etiology of ASD and the shared behavioral characteristics among family members took root with Folstein and Rutter's (1977) seminal article reporting higher concordance rates for ASD in monozygotic compared with same-sex dizygotic twins—a pattern that has been replicated in a number of subsequent twin studies (Bailey et al., 1995; Hallmayer et al., 2011; Le Couteur et al., 1996; Ritvo, Freeman, Mason-Brothers, Mo, & Ritvo, 1985). In addition, siblings have been found to have both an increased risk for ASD and elevated rates of autistic-like idiosyncrasies and communication disorders. For example, in the largest study of its day, Bolton et al. (1994) administered the Family History Schedule to 99 families of a child with ASD and found that as many as 20% of siblings and 11% of parents of diagnosed children with ASD exhibited subclinical autistic features. Based on this initial work, the study of family members of people with ASD garnered greater prominence in research, and the BAP emerged to denote a profile of milder characteristics of ASD in individuals who do not have an ASD diagnosis.

WHAT IS THE BROADER AUTISM PHENOTYPE?

Numerous studies have confirmed elevated rates of social, communication, personality, and cognitive characteristics that are associated with ASD in parents, siblings, and other extended family members of people with an ASD diagnosis (for reviews, see Bailey, Palferman, Heavey, & Le Couteur, 1998; Pisula & Ziegart-Sadowska, 2015; Sucksmith, Roth, & Hoekstra, 2011). For example, elevated difficulties in social skills in parents and siblings of children with an ASD are commonly found relative to parents and siblings of children with other developmental conditions, such as Down syndrome, while controlling for the stressor of a high-needs child in the home (Adolphs, Spezio, Parlier, & Piven, 2008; Bolton et al., 1994; Duchaine & Nakayama, 2006; Gillberg, 1989; Losh et al., 2009; Losh & Piven, 2007; Narayan, Moyes, & Wolff, 1990; Wolff et al., 1988). The presence of traits associated with ASD in parents and/or siblings has also been found to correlate with the severity of ASD in the child (Maxwell, Parish-Morris, & Hsin, 2013; Sasson, Lam, Parlier, Daniels, & Piven, 2013; Szatmari et al., 2008; Wilson, Freeman, Brock, Burton, & Palermo, 2010). For example, obsessive-compulsive traits in parents, particularly fathers, are correlated with children's repetitive behavior scores on the Autism Diagnostic Interview (Hollander, King, Delaney, Smith, & Silverman, 2003). This body of work suggests that the phenotypic variation associated with ASD may be due to underlying heritable genetic transmission—a possibility that has led to considerable interest in identifying biomarkers capable of explaining this phenotype (see Geschwind, 2011).

BAP characteristics are also present in people who do not have a child diagnosed with ASD. For example, in a study of more than 3,000 parents with and without a child with an ASD, Wheelwright, Auyeung, and Allison (2010) identified elevated BAP features in 33% of fathers and 23% of mothers of children with ASD, but also 22% of fathers and 9% of mothers of typically developing children. This type of sample that is made up only of parents likely represents an underestimation of the full BAP spectrum: Mate selection and the opportunities to pass on the phenotype to the next generation may be affected by some BAP features, such as those relating to social communication (Bailey et al., 1998).

A number of other disorders have been associated with ASD, including Rett's syndrome, Fragile X syndrome, and tuberous sclerosis (Geschwind, 2011; Rutter, 2005), congenital blindness, and disorders arising from profound institutional deprivation (Brown, Hobson, Lee, & Stevenson, 1997; Rutter, 2005; Rutter et al., 1999). However, these other disorders may represent phenocopies of ASD outcomes that resemble a genetic phenotype but do not arise from the same genetic cause (Goldschmidt, 1949). The BAP, too, may ultimately represent a phenocopy of ASD. However, given that ASD is currently diagnosed using behaviorally based methods and that the search for genetic causes continues, we suggest that the BAP can be a useful tool for studying the relationships among behaviors and traits that are similar to those in ASD.

Tools for Measuring the BAP

The first measures of the BAP were clinical interviews, developed to associate behavioral and genetic variability among members of a family with an individual with ASD in an attempt to understand the genetic underpinnings of the disorder (Bolton et al., 1994; Dawson et al., 2007; Piven et al., 1994). Because of the time required on the part of the participant, however, these clinical interviews are not widely used and have not been used to assess broad general population samples. Instead, brief surveys (some of which can be administered online) have been developed for this purpose. In the interest of introducing widely available BAP measures to the cognitive development community, we describe three here.

The Autism Spectrum Quotient (AQ) Questionnaire (Baron-Cohen, Wheelwright, Skinner, Martin, & Clubley, 2001) consists of 50 items asking participants to self-report the degree to which they agree with various statements concerning their strengths, weaknesses, and preferences along the domains of social abilities, communication, imagination, attention for detail, and attention switching. The scale is reported to have high test-retest reliability, internal consistency, sensitivity, and specificity (see Baron-Cohen et al., 2001). Versions of the AQ Questionnaire have been created for parents to complete about their adolescents (Baron-Cohen, Hoekstra, Knickmeyer, & Wheelwright, 2006) and children as young as 4 years of age (Auyeung, Baron-Cohen, Wheelwright, & Allison, 2007).

The Social Responsiveness Scale (SRS) (Constantino et al., 2003; Constantino & Todd, 2000) was originally designed as a reporting instrument to be completed by parents or teachers for children as young as 4 years old. A self-report version has since been developed for measuring the BAP in adults. The SRS contains 65 questions addressing the following five aspects of social behavior: social awareness, social cognition, social communication, social motivation, and autistic mannerisms. The SRS has high test-retest and interrater reliability (see Constantino et al., 2003; Constantino & Todd, 2003, 2005). Unlike the AQ, the SRS is

commercially produced and normed and was originally designed for clinical purposes. Specifically, the test was designed to identify the presence and severity of social impairment within the autism spectrum. It has good convergent validity with the Autism Diagnostic Interview during childhood (see Constantino & Gruber, 2012) but can also be a useful tool to measure subclinical variability across development.

The Broader Autism Phenotype Questionnaire (BAPQ; Hurley, Losh, Parlier, Reznick, & Piven, 2007) is a self-report measure suitable for adults. The BAPQ contains 36 statements and taps into the domains of aloofness, pragmatic language, and rigidity, which were intended to parallel the three *Diagnostic and Statistical Manual of Mental Disorders*-Fourth Edition ASD diagnostic domains of social impairment, language impairment, and repetitive behavior. The scale has good reliability and validity (see Hurley et al., 2007; Ingersoll, Hopwood, Wainer, & Donnellan, 2011; Sasson, Nowlin, & Pinkham, 2013) and also correlates well with the AQ and SRS (Ingersoll et al., 2011; Sasson, Nowlin, et al., 2013).

Additional tools that have been used in investigations of the BAP include the Children's Communication Checklist-2 (Bishop, 2003), the Pragmatic Rating Scale (Landa et al., 1992), and the Friendship Interview (Santangelo & Folstein, 1995). Unlike the surveys already described, these tools are more focused on the domains that they address. The Children's Communication Checklist-2 and Pragmatic Rating Scale provide detailed assessments of social communication skills in children aged 4 years and older, whereas the Friendship Inventory specifically measures an individual's interest in meaningful friendships. With such a full range of tools available, researchers interested in investigating the BAP as a correlate of a given skill should choose the tool best aligned with their research goals. For example, if a developmentalist was interested in investigating relationships among cognitive and social features of the BAP relative to children's academic performance, the AQ would provide the most breadth, with subscales addressing attention to detail and imagination in addition to communication and social skills. In contrast, if a developmentalist was interested in investigating social skills or communication to uncover nuances of the BAP as a correlate of school-based outcomes, the SRS provides four subscales to delineate social skills, while the Children's Communication Checklist provides four subscales for structural language and four subscales for pragmatic language.

As would be expected, individuals with an ASD diagnosis tend to score high on BAP measures. In a large meta-analysis, Ruzich, Allison, Smith, and Watson (2015) reported average scores of 35 on the AQ in people with an ASD, which is more than twice the general population's mean score of 17. Similarly, the SRS mean raw score for samples of individuals who have received an ASD diagnosis is greater than 80, while the mean raw score for typically developing participants is generally less than 40 (Constantino & Gruber, 2012). Thus, at any given age, individuals with and without ASD exhibit characteristically autistic-like behaviors at different frequencies.

Although the means between clinical and nonclinical samples differ, there is substantial overlap in the tails of the two distributions, and the BAP should not be used as a proxy for an ASD diagnosis as there are aspects of ASD that are qualitatively unique and not measured by the BAP. For example, a child with ASD may attend school with supports that set him or her apart from his or her classmates, and it may be clear to teachers and classmates that she or he is unable to participate fully independently. In contrast, a child with a high degree of BAP traits may exhibit enough strength in areas such as visuospatial perception or mathematics ability that

teachers and peers overlook any slightly awkward social behavior. A diagnosis of ASD entails a clinically significant impairment, which may not be present in some people who score high on a BAP measure.

Even if the BAP measures should not be used as proxies for an ASD diagnosis, one controversial question concerns whether they have any merit as screening tools. The creators of the AQ have suggested that a score greater than 32 is suggestive of an ASD and that one may consider further investigation in individuals with scores exceeding this value (Baron-Cohen et al., 2001), while the creators of the BAPQ are adamant that their tool should never be used for the purposes of screening (Piven & Sasson, 2014).

USING THE BAP TO STUDY ASD

In their meta-analysis, Ruzich et al. (2015) identified more than 850 articles that reported on the AQ—the measure most widely used in studies examining the associations between the BAP and performance on a number of different tasks tapping into a number of different constructs, including temperament (Pisula, Kawa, Danielewicz, & Pisula, 2015), language abilities (Whitehouse, Barry, & Bishop, 2007), sensory sensitivity (Robertson & Simmons, 2013), and visuomotor associative learning (Parkington, Clements, Landry, & Chouinard, 2015). We believe that these kinds of efforts with nonclinical samples can contribute to the study of ASD. For example, a particular skill pattern that is unique to those who have received a diagnosis of ASD and is not related to BAP scores among those who do not have a diagnosis could represent a key to the causative model of the disorder or an area of functioning that has been severely impacted by the disorder. This finding is seen with IQ, as low IQ is found at a much higher rate among persons with ASD but does not appear to be related to scores on the BAP (Szatmari et al., 1995). One explanation could be that ASD may involve problems in brain development that interfere with fundamental perceptual and learning skills, thereby limiting children's measurable IQ. Additionally, the communication abilities necessary for valid testing may also be uniquely compromised by ASD. Or imagine siblings who both receive an elevated score on a BAP measure, but only one of them has received a diagnosis of ASD. Exploring qualitative and quantitative differences between these individuals may help us to understand what contributes uniquely to ASD.

One critique of the use of the BAP to study ASD is that typically developing individuals have qualitatively different experiences from those who have received a diagnosis of ASD. How can we expect a model based on quantitative variability in the magnitude of symptoms to reflect the qualitative difference between growing up with or without a clinical diagnosis? If researchers employing the normal-range analogue model have not included participants with clinical impairment, how can this model inform our understanding of the clinical disorder? These are good questions, but in our perspective, using the BAP to understand ASD is somewhat analogous to using animal models to understand human behavior. The BAP complements, but does not substitute for, research conducted with individuals who have received a diagnosis. It provides complementary perspectives on questions and foundational work for future studies in samples with ASD.

Five Benefits of the BAP Model

Indeed, in our view, using the BAP as a model to understand ASD has several potential benefits. One, given that the BAP can be measured in nonclinical samples, it is possible to conduct

studies with many more participants than in most studies that focus on individuals with ASD. Additionally, because the BAP involves the full spectrum, it offers a great range of variability, which in turn allows for correlation-based analyses, which have far greater statistical power than between-group analyses (Mitchell & Jolley, 2013, p. 271). For example, Chouinard, Unwin, Landry, and Sperandio (2016) demonstrated significant correlations between AQ scores and reduced susceptibility to visual illusions, while a between-group analysis on the same data comparing “high” and “low” AQ groups along a median split failed to arrive at the same result, which the authors attributed to the former analysis offering greater statistical power for explaining the variability in the data compared with the latter.

Two, using the BAP can allow for the investigation of questions that might be practically difficult to study among persons with ASD. For example, many cognitive studies require prolonged testing sessions and multiple experiments to precisely delineate and understand specific cognitive processes. Imaging studies may require participants to endure uncomfortable sensory experiences, including remaining still in a functional magnetic resonance imaging scanner or wearing a tight electroencephalogram cap. In some cases, these experimental requirements can be especially challenging for children and adults with ASD. Using the BAP can allow researchers to recruit participants who can take part in multiple experiments and to compare their performance across those experiments.

Three, comorbidity can be more easily controlled for in typically developing participants, as other disorders (e.g., seizures, attention-deficit hyperactivity disorder [ADHD], and general anxiety disorder) are also frequently present among persons with ASD (Simonoff et al., 2008). For example, a person with an ASD may also manifest ADHD. It then follows that any cognitive difference compared with a typically developing person may be the result of ASD and/or ADHD. Using the BAP as a model provides an alternative to either excluding individuals with ASD with a comorbid disorder or having to collect multiple cohorts of other disorders as a way to control for comorbidity.

Four, using the BAP can allow researchers to more easily control for both chronological and mental ages during development. Less than 50% of people with an ASD have average or above-average intelligence, and at least 30% are considered to have an intellectual disability (Centers for Disease Control and Prevention, 2014; Elsabbagh et al., 2012; but see Courchesne, Meilleur, Poulin-Lord, Dawson, & Soulières, 2015). Many cognitive abilities are influenced by both chronological and mental age, yet most studies match the control group to the ASD group based on either mental or chronological age because of intrinsic difficulties in matching for both. The use of the BAP model in typically developing children does not have this limitation and allows for the opportunity to control for both chronological and mental ages and consequently to understand how a cognitive skill develops over the course of typical development and how this trajectory is associated with BAP features.

Five, using the BAP as a model provides the opportunity to examine the developmental sequence and correlates of skills implicated in ASD in greater isolation. By definition, an ASD diagnosis includes multiple symptoms from both the social communication and repetitive behavior domains. In contrast, individuals can have elevated BAP scores in single domains—for example, aloof behavior without elevated scores in pragmatic language or rigid behavior domains—providing opportunities to examine how delineation of BAP features in isolation might relate to ASD. This is relevant to the discussion of the inherent links in ASD among the classic triad of impairments in social interaction, communication, and restricted and repetitive behaviors. For example, based on convergent evidence from twin studies showing heritability within a domain of the triad but not

across domains as well as patterns of correlated and uncorrelated skill sets and impairment associated with ASD, Happé and colleagues (Happé & Ronald, 2008; see also Brunsdon & Happé, 2014; Happé, Ronald, & Plomin, 2006) have argued that there is no single cause for impairments in these three domains, but rather three uncorrelated areas of impairment with distinct genetic causes. These “fractionated” domains can be ideally tested within the BAP model by examining patterns of high and low BAP features.

USING THE BAP TO STUDY DEVELOPMENT

We have so far concentrated our discussion on how the BAP can be used to study ASD—what is unique to clinical samples and what is shared with nonclinical ones. However, the BAP can also contribute to understanding the developmental trajectories of autistic tendencies. For example, infants who have an older sibling with an ASD diagnosis are more likely to receive a diagnosis of ASD than are infants who do not have a sibling with an ASD diagnosis (10% vs. 1%; Constantino, Zhang, Frazier, Abbacchi, & Law, 2010). There has thus been considerable interest in studying these infant siblings as a means to investigate longitudinally the earliest-emerging signs of ASD (for a review, see Jones, Gliga, Bedford, Charman, & Johnson, 2014).

One area that has received a good deal of study in infant sibling research involves visual attention: Infants who have a sibling with ASD and who are later diagnosed with ASD show different patterns of visual attention from infants who do not have a sibling with ASD and who are not later diagnosed with ASD (Elsabbagh et al., 2013; Zwaigenbaum et al., 2005). Several researchers have suggested that these early atypical patterns of visual attention are a part of a cascade of events that may contribute to a later diagnosis of ASD (Landry, Mitchell, & Burack, 2009; Sacrey, Armstrong, Bryson, & Zwaigenbaum, 2014). These findings can also be nuanced. For example, infants who have a sibling with ASD and who are not later diagnosed with ASD themselves also show atypical patterns of visual attention, but they are not as extreme as the patterns of those who are later diagnosed with ASD (Elsabbagh et al., 2009). To better understand the developmental outcomes of all these children, longitudinal research will provide insight into how these patterns of early visual attention map on to later BAP profiles, particularly as visual perception is a skill that varies in adults as a function of the BAP (Almeida, Dickinson, Maybery, Badcock, & Badcock, 2010; Chouinard, Noulty, Sperandio, & Landry, 2013; Chouinard et al., 2016; Grinter, Maybery, et al., 2009, Grinter, Van Beek, et al., 2009; Sutherland & Crewther, 2010; Walter, Dassonville, & Bochsler, 2009).

The BAP model can also contribute to our understanding of development by allowing for the investigation of how skills interact across development—an approach that may be particularly useful for examining skills that typically emerge around the same time in typical development. For example, during the preschool period, typically developing children make tremendous gains in both executive functioning (EF) and theory of mind (ToM). Given the timing of both skills in typical development, combined with evidence of impairments of both among persons with ASD, there has been a good deal of debate about the nature of the relationship between them (reviewed in Brunsdon & Happé, 2014): Is EF required for ToM (e.g., Pellicano, 2007, 2010), or are the two more or less independent? It remains unclear whether children who score high on BAP measures show patterns of performance on EF and/or ToM tasks similar to children with ASD (Sucksmith et al., 2011). Studies that incorporate the BAP framework may be able to contribute

to this debate, which is relevant to theories of ASD but also to theories of typical development. In particular, the BAP model offers the opportunity to examine skills, not only categorically as impaired and unimpaired but along a fully continuous spectrum.

CONCLUSIONS

Imagine baking a cake. There are many ingredients involved and seemingly infinite variations that still lead to a functional and delicious dessert. Any single ingredient can also be manipulated within certain limits without compromising the ultimate integrity of the end product, but too much or too little can lead to problems. We have spent decades looking for the one ingredient that explains autism, but it has not been a fruitful approach—it appears to be more analogous to the little variations, with a pinch too little and/or too much of several ingredients in combination, and our best approach to understand how the ingredients work together is to examine the full range of human variability. The BAP model was born out of recognizing the range of variability in the typically developing population in some of the same domains that are atypical in ASD. Expanding the BAP model into the general population will help us in our quest to uncover these variations, their developmental trajectories, those that might be unique to ASD, and what makes them unique.

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